Band-1 Hadamard Order 4036 — Miyamoto (Double-seed H_1008) + Paley-II; Gap 2052 Still OPEN

H_504/H_1008/C_1010/H_4036 PASS; 4036 in SPK via Paley-II; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_3908_Miyamoto`; `2026-09-15_Band1_Hadamard_3748_Miyamoto` (Miyamoto + Paley-II SPK pattern).
Lab. `/workspace/experiments/band1/hadamard_4036_miyamoto/`.
Cutoff. 2026-09-15T13:27:29Z (~9:27AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 504, 1008, and 4036 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double-seed H_{1008} = Sylvester-double(H_{504}) where H_{504} is Paley type I ($q = 503$), and independently via Paley type II. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4036 **enters SPK via Paley-II** ($q = 2017$ prime $\equiv 1 \pmod 4$; order $2(q+1) = 4036$): Paley-I **FAIL** ($q = 4035$ not prime); Kronecker **FAIL** ($4036 = 4 \cdot 1009$, 1009 not Hadamard-admissible). Seed H_{1008} uses Sylvester double of Paley-I ($q = 503$); Paley-II seed for 1008 **FAIL** ($q = 503 \equiv 3 \pmod 4$). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 3908 on box. This increment ships **order 4036** ($4036 = 4 \cdot 1009$, 1009 prime $\equiv 1 \pmod 4$) via the **Double-seed** pattern plus a classical **Paley-II SPK path** (mirroring 3748 Double-seed + Paley-II). Seed H_{504} via Paley type I ($q = 503$); H_{1008} = Sylvester-double(H_{504}); symmetric Paley conference C_{1010} over prime 1009; H_{4036} = Miyamoto(H_{1008}, C_{1010}). Independently, H_{4036} via Paley type II ($q = 2017$). Unit tests **28/28 PASS**. SPK next gap remains **2052** (OPEN). Order 4036 **is in SPK** via Paley-II (not Miyamoto-only). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4036; Miyamoto; Sylvester doubling; Paley type I; Paley type II; Paley conference; constructive gap 2052; SPK; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{504} constructed and verified (Paley type I, $q = 503$). 2. H_{1008} constructed and verified (Sylvester double of H_{504}). Paley-II seed for 1008 **rejected**. 3. C_{1010} constructed and verified (Paley conference, $q = 1009$): $CC^\top = 1009I$, zero diagonal. 4. H_{4036} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. H_{4036} also constructed and verified (Paley type II, $q = 2017$) — SPK classical path. 6. SPK: $4036 \in$ SPK via Paley-II. Paley-I/Kronecker paths for 4036 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$4036 = 4 \cdot 1009$
$q = 1009$	prime, $1009 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1008}
Needed conference	order $1010 = q + 1$
H_{504} path	Paley type I, $q = 503$ (prime $\equiv 3 \pmod 4$)
H_{1008} path (used)	Sylvester double of H_{504}
H_{1008} Paley-II	**REJECTED**: $2(q' + 1) = 1008 \Rightarrow q' = 503$ prime but $\equiv 3 \pmod 4$
Conference path	Symmetric Paley over prime 1009
Paley-I for 4036	**REJECTED**: $q' = 4035$ not prime
Paley-II for 4036	**PASS**: $q' = 2017$ prime $\equiv 1 \pmod 4$; $2(q' + 1) = 4036$
Kronecker for 4036	**REJECTED**: only 4×1009; $1009 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{504}	Paley type I, $q = 503$	0	0.2023	**PASS**
H_{1008}	Sylvester double of H_{504}	0	0.4695	**PASS**
C_{1010}	Paley conference, $q = 1009$	0	0.9385	**PASS**
H_{4036}	Miyamoto(H_{1008}, C_{1010})	0	27.7065	**PASS**
H_{4036}	Paley type II, $q = 2017$	0	30.4352	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh + Paley-II) $\approx$ 61.8371 s. Compressed ± 1 matrices saved as `H504_pm1.npz`, `H1008_pm1.npz`, `C1010_pm1.npz`, `H4036_pm1.npz`, `H4036_paleyII_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior classical/SPK ships (through 3908/3748/976/488/936/468/3316/3188/ **exclude** 3716/3524/3284/3076/3044/2836/2692/2564 Miyamoto-only) $\cup$ verified H_{504}, $H_{1008} \cup$ Paley-II order 4036 $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4500). **Miyamoto is not folded into SPK**; order 4036 enters SPK by the classical Paley-II path.

Quantity	Value
Known count	964
4036 in SPK	**True** (via Paley-II)
1008 in SPK	**True** (Sylvester double of Paley-I)
504 in SPK	**True** (Paley-I)
Next constructive gap	**2052** (OPEN)
Miyamoto-only bonus?	**No** (in SPK via Paley-II; like 3748)

5. Gap 2052 (OPEN; method triage / non-finding ONLY)

Path	Result
Paley-I ($q = 2051$)	FAIL
Paley-II ($q = 1025$)	FAIL
Kronecker	FAIL ($2052 = 2^2 \cdot 3^3 \cdot 19$)
Miyamoto + Paley conference ($q = 513$)	FAIL (513 not prime)
GS / SDS / Williamson / Yamada	lit-only; not constructed this run

Status: OPEN. Non-finding $\neq$ counterexample. Conjecture remains OPEN.

6. Unit tests

28/28 PASS (Sylvester/Paley/Miyamoto controls; $q = 503/1009/2017$ arithmetic; Double path for H_{1008}; Paley-II seed path for 1008 rejected; SPK path eligibility for 4036 via Paley-II; reject-all-ones; conference control; gap-2052-not-filled guard).

7. Non-claims (explicit)

- No full Hadamard conjecture claim.
- No Clay-closed stamp; `clay_closed = false`.
- No finished-ToE claim; GATE-04 **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims.
- Gap **2052** remains **OPEN** under SPK.
- Order 4036 is **in SPK via Paley-II** (Miyamoto also verified; not Miyamoto-only).

8. Next optional

method-attack SPK gap **2052** OR next Miyamoto ($q = 1049 \to n = 4196$ Double-seedable) OR Track B privacy / Object 1 meter await.

Lab: `/workspace/experiments/band1/hadamard_4036_miyamoto/` · *Script:* `verify_hadamard_4036.py` · *Artifacts:* `results.json`, `SUMMARY.txt`, `literature_ledger.json`, `*.npz`.